# Advanced Treatment Options

**A Guide to Deep Brain Stimulation, Pump Medications & Focused Ultrasound**

Parkinson's Foundation

## About This Guide

As Parkinson's disease (PD) progresses over time, managing movement symptoms with standard medications can become more challenging. When this happens, people often explore other treatment options. These include deep brain stimulation (DBS), pump medications and procedures such as focused ultrasound.

This guide explains how these treatments work, why they are used and what to know before or after starting them. It aims to answer common questions and support informed conversations with your care team.

The information in this guide is not meant to replace medical advice, diagnosis or treatment. Always talk with your care team about your symptoms, treatments and questions.

### Parkinson's Foundation Resources

Explore our books, fact sheets, webinars, podcasts and more in our PD Library at Parkinson.org/Library.

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# 1

# Exploring Advanced Treatment Options

Parkinson's disease (PD) causes movement symptoms such as slowness, stiffness, tremor and balance problems, along with a range of non-movement symptoms including fatigue, mood changes, sleep issues, speech and swallowing difficulties and others.

Although there is no cure for PD at this time, treatments can help manage symptoms and improve quality of life. Medications often control movement symptoms for years, but as the disease progresses, their effects may become less predictable and side effects harder to manage. For some, this may be a sign to explore advanced treatment options.

### In this chapter, you'll learn about:

- Dopamine loss and Parkinson's
- Medications and symptom control
- Advanced treatment options

# Dopamine Loss and Parkinson's

Dopamine is a brain chemical that helps control movement, mood, motivation and other important functions. In Parkinson's, the cells that make dopamine get damaged and die over time. As a result, the amount of dopamine in the brain slowly decreases.

# Understanding Parkinson's Medications

Treatment for Parkinson's often includes medications, exercise and other strategies to manage symptoms. Medications play a key role, especially for movement symptoms. Many people with PD take medications that boost, mimic or replace dopamine.

Two main types are commonly used:

- **Levodopa**, which the brain converts into dopamine, is the most effective treatment for movement symptoms.
- **Dopamine agonists**, act like dopamine and may be used alone or with levodopa.

Movement medications are tailored to each person and adjusted as symptoms change, or side effects appear. Other drugs may help medications last longer, reduce side effects or manage non-movement symptoms.

Learn more at Parkinson.org/MedicationsBook.

## Levodopa

Levodopa has been used to treat Parkinson's disease since the 1960s. It remains the most effective and commonly prescribed medication for movement symptoms because it is the only one that directly replaces dopamine.

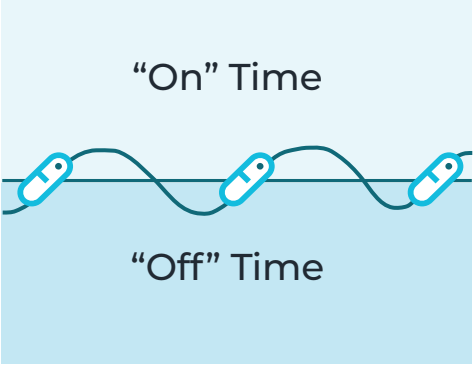
It comes in several forms, including tablets, capsules and other options. Many people start with immediate-release tablets but may switch to a different form or combine multiple types of levodopa — along with other medications — to help manage symptoms.

## Movement Fluctuations and Medication Timing

In early Parkinson's, taking levodopa three times a day often provides steady symptom control. The brain still has enough dopamine-producing cells to store and release each dose gradually.

As the number of these cells decreases, the brain becomes less able to regulate dopamine between doses. This can lead to shorter and less predictable periods of symptom improvement — a pattern known as movement fluctuations. People may experience:

- **"On" time:** When the medication is working and symptoms improve. Dyskinesia (extra, involuntary movements), a possible side effect of levodopa, may or may not be present.
- **"Off" time:** When the medication wears off and symptoms return. For some people, dyskinesia happens during "off" periods.




## How Food and Digestion Affect Levodopa

Most forms of levodopa enter the bloodstream through the small intestine, where protein from foods like meat and dairy can interfere. High-fat meals don't compete directly, but may slow digestion which can delay absorption.

Constipation or delayed stomach emptying (gastroparesis) can affect medication absorption. As dosing increases to manage symptoms, timing meals around medication becomes more complex.

### What is the therapeutic window?

The therapeutic window is the period during "on" time when Parkinson's medication works well without causing side effects. As Parkinson's progresses, this window often gets shorter, which may mean more medication adjustments.

### Difficult-to-Treat Movement Symptoms

Some movement symptoms, such as tremor, may not respond well to medication, even when other symptoms improve. This may be because tremor involves brain areas less sensitive to dopamine. Dystonia — painful muscle cramping or twisting — can also be difficult to manage. It may result from low dopamine levels or as a side effect of Parkinson's medications, especially levodopa.

## Common Medication Adjustments

Medications often need updates as symptoms change to help smooth out symptom control and reduce side effects. Finding the right balance can take time and trial and error. It is important to work closely with a neurologist experienced in Parkinson's.

Adjustments may include:

- Increasing or decreasing the levodopa dose size or frequency
- Switching to a longer-acting form of medication
- Adding another medication, such as a dopamine agonist, COMT inhibitor or MAO-B inhibitor
- Taking an on-demand medication to manage unexpected "wearing off" or as needed
- Adding amantadine to help treat dyskinesia

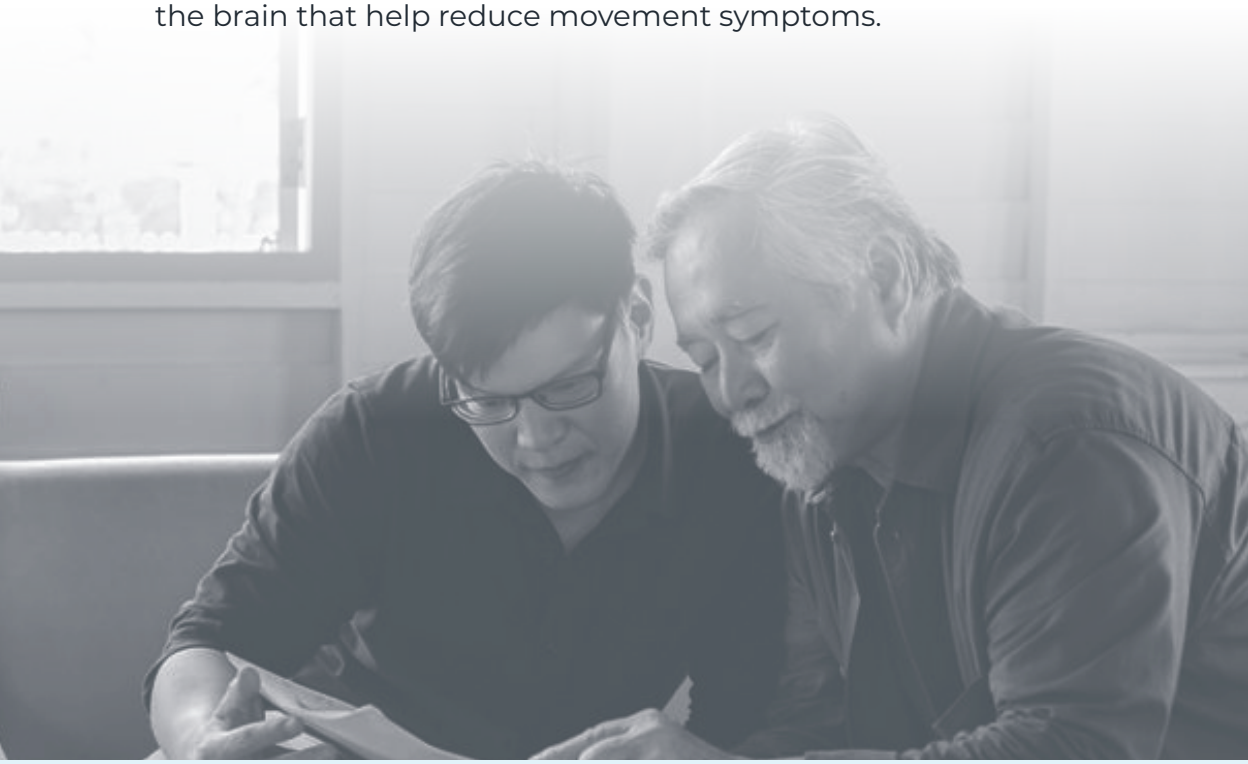
Read the Managing "Off" Time fact sheet at Parkinson.org/Library to learn more.

# Advanced Treatment Options

When medication adjustments aren't enough to manage movement symptoms, other treatment options may help. You can explore these with your care team to find what's right for you. These approaches may improve symptom control, but at this time, no treatment can slow the progression of Parkinson's.

The advanced treatment options discussed in this guide include:

- **Pump medications:** A small pump delivers medication through a tube under the skin or directly into the intestine. Examples include Duopa, Onapgo and Vyalev.
- **Deep brain stimulation (DBS):** A surgical procedure that places thin leads (wires) in the brain to deliver electrical signals that help control movement symptoms.
- **Focused ultrasound and other lesion therapies:** These procedures use targeted energy to create small cuts or lesions in the brain that help reduce movement symptoms.



# Steps Before Advanced Treatments

Before moving forward with advanced treatments, consider these important steps:

- ☐ **Make sure advanced treatments are right for you.**
  Confirm your diagnosis and be sure you have fully explored medication adjustments. Consider getting a second opinion from a movement disorders specialist (a neurologist with additional training in PD).
- ☐ **Understand your options.**
  Ask about the risks, benefits and what the process involves for each treatment.
- ☐ **Learn from others.**
  Connect with people who use advanced treatments to hear about their experiences.
- ☐ **Choose an experienced treatment center.**
  Look for a center with the expertise, equipment and support needed for advanced treatments such as deep brain stimulation. Find a center through the Parkinson's Foundation Global Care Network Centers at Parkinson.org/GlobalCareNetwork.
- ☐ **Talk openly with your care team about your treatment goals.**
  Make sure your expectations are realistic and in line with what the treatment can actually do.
- ☐ **Consider the impact on your life.**
  Think about how the treatment will affect your routine. For example, will it require frequent follow-up visits or limit your activities?
- ☐ **Evaluate your support system.**
  Will family, friends or care partners be available to help with treatment, recovery and ongoing care?
- ☐ **Plan for costs.**
  Understand out-of-pocket expenses and insurance coverage to avoid surprises.

## Not sure where to start?

Reach out to the Parkinson's Foundation Helpline to learn more about treatment options.

Our Parkinson's information specialists can share details about DBS and other advanced therapies and help you explore resources in your area.

Call **1-800-4PD-INFO (1-800-473-4636)** or email **Helpline@Parkinson.org**.

## Looking for insight?

My PD Story features real stories from people in the Parkinson's community. Hear from those who have chosen DBS or other advanced treatments. Scan the QR code or visit Parkinson.org/Story.

### Share Your Story

Want to share your journey with Parkinson's?
Visit Parkinson.org/MyStory.

# 2

# Pump Medications

For some people living with Parkinson's disease (PD), symptoms may continue to be difficult to manage even after adjusting medications. Side effects may also occur at lower doses. In these cases, pump medications — also called "pump therapies" — may offer another option.

These therapies release a steady flow of medication during the day — and sometimes at night — which can help reduce "off" time and lessen medication-related side effects. Pump therapies may be considered when oral medications no longer work well, even after adjusting timing, dosage or adding other drugs.

**In this chapter, you'll learn about:**

- How pump therapies work
- Why pump therapies are used
- What to expect from each type

# Overview of Pump Therapies

Pump therapies use a small, wearable device to deliver Parkinson's medications continuously through a thin tube placed either just under the skin or directly into the small intestine. The two medications used are levodopa and apomorphine (a dopamine agonist). These treatments mainly help manage movement symptoms, such as tremor, stiffness and slow movement.

# Benefits of Continuous Medication Delivery

Pump therapies release medication continuously throughout the day — and sometimes at night. This type of delivery can help maintain more stable dopamine levels and reduce "off" time. It also allows for lower, more steady medication amounts, which can lessen side effects.

These therapies don't move through the digestive tract like pills. This helps improve medication absorption and eliminates the need to time doses around protein-rich meals (which can interfere with levodopa absorption for some people).

For some people, not having to manage pills multiple times a day is a welcome change. Pump therapies do require ongoing maintenance — such as switching the infusion site (or insertion site) and replacing the medication every day or every few days — depending on the therapy.

To review the advantages and considerations of advanced therapies, see **Chapter 1: Exploring Advanced Treatment Options**.

Other advanced treatments, such as deep brain stimulation (DBS), may also be an option. The right choice for you will depend on many factors including symptoms, overall health, personal preference and insurance coverage.

> I started on Sinemet three or four times a day — and it worked okay at first.
>
> As time went on, the effects didn't last as long and timing my meds around meals got trickier.
>
> So far, the pump has helped me keep my symptoms more under control and made daily life easier.
>
> ***— Luke, person with PD***

## Who might be a candidate for pump therapies?

Pump therapies might be an option for you if you:

- Have movement symptoms or significant "off" time that standard medications no longer control well
- Experience side effects such as dyskinesia even at lower doses
- Have trouble absorbing medication due to constipation, slow stomach emptying or because protein in your diet affects how the medication works
- Can manage the pump system at home or have support to help if needed

If you're curious whether pump therapies could work for you, talk to your doctor.

# Types of Pump Therapies

Three pump therapies are approved by the U.S. Food and Drug Administration (FDA) for movement fluctuations in advanced Parkinson's disease.

All three systems are wearable. Depending on the system, the pump can be clipped to a belt, worn under clothing in a vest or hung around the neck with a strap. The Vyalev system uses rechargeable batteries and comes with a spare. Duopa and Onapgo use regular batteries that need to be replaced when they run low.

## Duopa System

Carbidopa/levodopa enteral suspension (Duopa) was approved by the FDA in 2015 for managing "off" time in advanced Parkinson's disease.

### Components & How They Work:

- **PEG-J tube:** This is a tube that is surgically placed through a small opening in the stomach wall to deliver the medication directly into the small intestine.
- **Cassette:** This contains the carbidopa/levodopa suspension and is inserted into the pump each day.
- **Pump:** This wearable device is about the size of two large cellphones. It delivers medication continuously for about 16 hours and is programmed with a morning dose and optional extra doses.
- **Backup plan:** If the system is disconnected for more than two hours, oral medications may be needed until the pump is restarted.

## Vyalev and Onapgo Systems

Foscarbidopa/foslevodopa (Vyalev) and apomorphine hydrochloride (Onapgo), both for subcutaneous infusion, were approved by the FDA in 2024 and 2025, respectively, to help manage movement fluctuations in advanced Parkinson's disease.

### Components & How They Work:

- **Medication container:**
  - Vyalev comes in a vial and is drawn into a syringe.
  - Onapgo comes in a prefilled cartridge.
- **Pump:** This wearable device is about the size of one or two cellphones.
  - For Vyalev, the syringe is placed in the pump, which delivers a steady amount of medication over 24 hours. Optional changes to the delivery rate can be programmed to help manage symptoms.
  - For Onapgo, the cartridge is inserted into the pump, which delivers medication over 16 waking hours. If programmed, extra doses can be given when needed.
- **Cannula:** This is a small, flexible tube that sits just under the skin to deliver medication.
  - It is inserted using a small needle, most commonly in the abdomen, and is replaced every few days.
  - The infusion site (where the tube enters the skin) should be changed regularly to prevent irritation and infection.
- **Tubing:** This connects the pump to the cannula and is replaced every few days along with the cannula.
- **Backup plan:** If the system is disconnected for more than an hour, the cannula and tubing may need to be replaced, and oral medications are typically used until the pump is restarted.

# Pump Medication Information

| Name | Form & Strength | Typical Treatment |
|---|---|---|
| Carbidopa/ levodopa enteral suspension (Duopa) | 4.63 milligrams (mg) carbidopa and 20 mg levodopa per milliliter (mL); in a medication cassette | Up to 2,000 mg of levodopa delivered over 16 hours |
| Foscarbidopa/ foslevodopa (Vyalev) for subcutaneous infusion | 120 mg foscarbidopa and 2,400 mg of foslevodopa per 10 mL vial | Dose individualized; typically up to 3,525 mg foslevodopa (about 2,500 mg levodopa) over 16 to 24 hours |
| Apomorphine hydrochloride (Onapgo) for subcutaneous infusion | 98 mg/20 mL single-dose cartridges | Maximum total daily dose is 98 mg, spread throughout waking hours; anti-nausea medications may be needed |

### Medication Adjustments

Finding the right dose and rate for pump medications can take time. The care team may make adjustments to help manage symptoms and side effects. Noticing and sharing any changes can help guide treatment.

## Side Effects

Side effects can vary from person to person and may be influenced by factors such as age, sex, weight, dose, individual sensitivity and where you are in your Parkinson's journey. This is not a complete list of possible side effects — talk with your care team about any concerns or questions.

### Common Levodopa Side Effects

- Nausea
- Low blood pressure (may cause dizziness)
- Dyskinesia
- Hallucinations (more common in advanced stages of PD)

### Common Dopamine Agonist Side Effects

- Nausea
- Low blood pressure (may cause dizziness)
- Excessive daytime sleepiness
- Leg swelling
- Hallucinations
- Compulsive behaviors (such as intense, uncontrolled urges to shop, gamble, eat or engage in sexual activity) — important to catch early and make treatment adjustments

### Side Effects Specific to Pump Therapies

- Site irritation or infection may occur where the medication is delivered.
- For carbidopa/levodopa enteral suspension (Duopa), there's a risk of infection at the stoma site (where the tube enters the stomach).

**Q How much do pump therapies cost?**

**A** Insurance coverage for pump therapies varies depending on your plan. Some treatments can be very expensive and may not be fully covered, especially newer ones. Manufacturers often offer assistance programs, but these can have limits or time restrictions. It's important to talk with your healthcare team about costs and available support to help plan ahead.

**Q Can I shower or bathe while using pump medications?**

**A** Pump therapy routines vary. Vyalev and Onapgo pumps can be removed for showering or swimming. The Duopa pump is connected internally and usually stays in place, so it's covered to keep it dry during bathing.

**Q Do I sleep with the pump at night?**

**A** It depends on the therapy — some pumps run overnight. Others are disconnected at bedtime. Your care team will guide you based on your specific treatment.

**Q Does inserting the cannula hurt?**

**A** You might feel a brief pinch or burning sensation when the cannula is inserted under the skin with the needle. This usually lasts only a few minutes.

**Q How do I get the medication and how is it stored?**

**A** Medication is typically shipped to your home or doctor's office and kept refrigerated. Your healthcare team will guide you on safe storage and handling.

**Q Who can help me manage the device at home?**

**A** Support from family, care partners or home health services can be helpful, especially at first. Your care team can connect you to additional resources.

# Support for Pump Therapies

Each manufacturer offers education and device support for its pump therapy system. Information is available on the manufacturer's website. After a prescription for a specific pump therapy, people often qualify for one-on-one support programs. Your care team can connect you to these services and point you to any financial programs that may help with the cost.

## Education & Resources

**Duopa** (AbbVie): Duopa.com

**Vyalev** (AbbVie): Vyalev.com

**Onapgo** (Supernus Pharmaceuticals): Onapgo.com

## Device Support Programs

**Duopa support (DuoConnect Complete):**

- Phone: 1-844-438-6720 (8 a.m. – 5 p.m. ET, Monday – Friday)
- After-hours: 1-844-DUO4YOU (1-844-386-4968)

**Vyalev support (Vyalev Complete)**

- Phone: 1-866-489-2538 (24/7)

**Onapgo support (Circle of Care Program)**

- Phone: 1-833-3-ONAPGO (1-833-366-2746)
  (8 a.m. – 5 p.m. ET, Monday – Friday)

# 3

# An Overview of DBS

Deep brain stimulation (DBS) is a type of brain surgery in which thin wires are placed into specific areas inside the brain through a small opening in the skull. These wires connect to a small device implanted under the skin near the collarbone. The system sends electrical signals that can help reduce movement symptoms such as tremor, stiffness and slowness, and may also improve certain medication-related challenges.

Similar to other advanced treatments, DBS may be considered when oral medications no longer control symptoms or cause difficult side effects. Because it involves brain surgery, the decision to pursue DBS requires careful evaluation and planning with your care team.

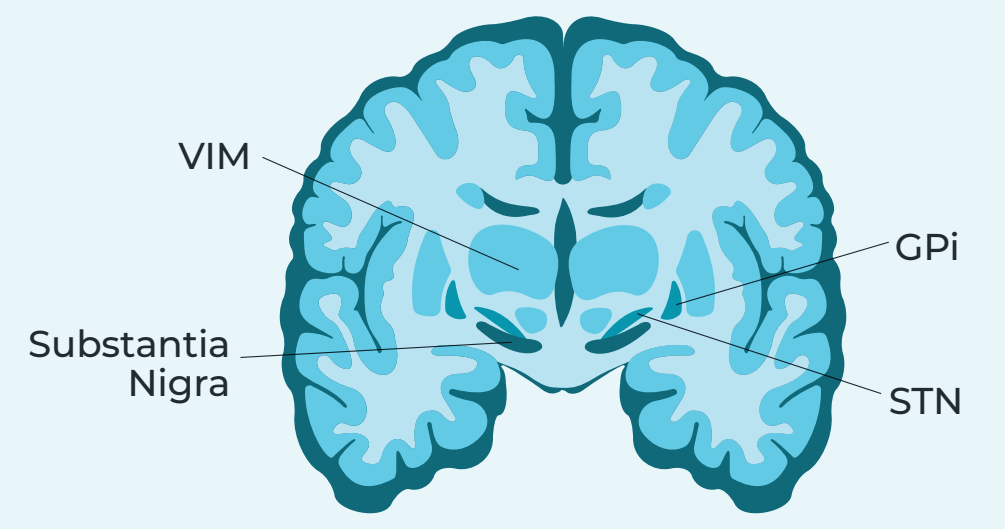
**In this chapter, you'll learn about:**

- The science behind DBS
- Symptoms DBS treats
- The components of the DBS system
- Choosing a DBS system

# The Science Behind DBS

Parkinson's affects a part of the brain called the **substantia nigra**. This area has more than 400,000 cells that produce dopamine, a chemical that helps control movement, mood and other functions. In Parkinson's, these cells are damaged and die over time.

The substantia nigra connects to other areas of the brain, including the globus pallidus internus **(GPi)** and the subthalamic nucleus **(STN)**. These regions control movement in the arms, legs and neck, and also influence mood, thinking and more.




As dopamine levels gradually decrease, brain regions involved in movement stop working together smoothly. This leads to abnormal brain activity, which can cause symptoms such as tremor, stiffness and slowness.

DBS sends electrical pulses through thin wires to help reset abnormal brain signals and improve movement. It usually targets the GPi or STN, and sometimes the ventral intermediate nucleus **(VIM)** in the thalamus. While the exact reason it works isn't fully understood, DBS has been shown to reduce movement symptoms.

# Symptoms DBS Helps Manage

DBS was approved by the U.S. Food and Drug Administration (FDA) for advanced Parkinson's in 2002. Over time, approval has expanded to include earlier stages and additional features. While DBS doesn't slow Parkinson's progression, it's considered a safe and effective treatment for people who are good candidates.

DBS is mainly used to improve movement symptoms and help smooth fluctuations. It may also help manage some non-movement symptoms. In general, symptoms that respond to levodopa tend to respond well to DBS. After the procedure, many people find they can take less medication, which often helps reduce side effects.

While DBS can improve symptoms, it is not a one-time procedure. Medications and stimulation settings require ongoing adjustment.

### DBS may help with:

- **Movement symptoms** such as tremor, stiffness and slow movement, especially if they respond to levodopa. Tremor can improve with DBS even if it doesn't respond to levodopa.
- **Medication side effects**, such as dyskinesia (extra, involuntary movements), directly or by allowing lower medication doses.
- **Some non-movement symptoms** including sleep issues, pain and overactive bladder.

### DBS may not help:

- **Mood, energy and walking**, especially if these didn't improve with levodopa before DBS.

### DBS may worsen:

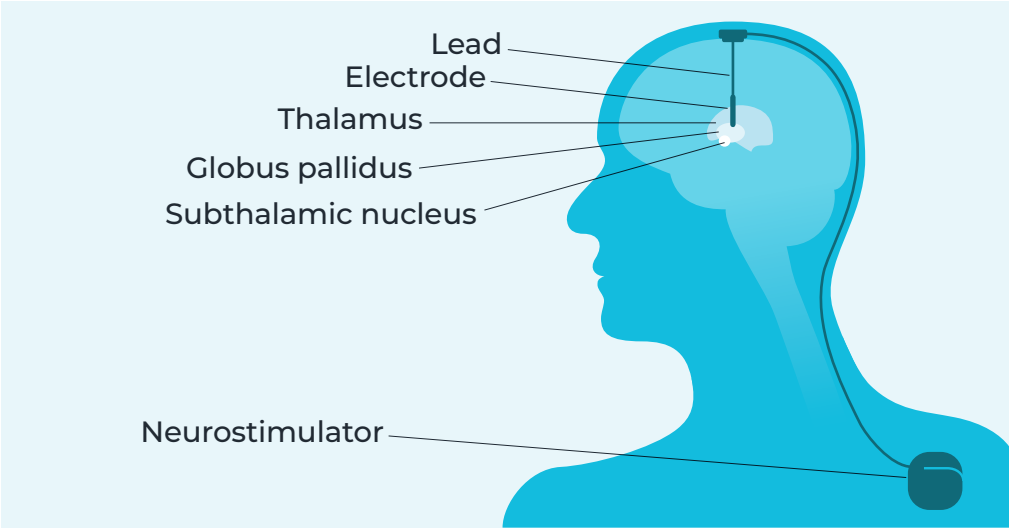
- **Balance, speech and swallowing problems**, especially if experienced before DBS. Typically, these changes are related to stimulation and can improve or go away by adjusting the device settings.

# Brain Areas for DBS

DBS helps manage Parkinson's symptoms through delivering electrical stimulation to specific areas of the brain. The FDA has approved three target areas for DBS. The table below explains the benefits of each one.

| DBS Target Areas | Stimulation Effects |
|---|---|
| **Globus pallidus internus (GPi)** | **Reduces:**<br>• Tremor<br>• Stiffness (rigidity)<br>• Slow movement (bradykinesia)<br>• Dystonia<br>• Dyskinesia<br>• Movement fluctuations |
| **Subthalamic nucleus (STN)** | **Reduces:**<br>• Tremor<br>• Stiffness (rigidity)<br>• Slow movement (bradykinesia)<br>• Movement fluctuations<br>STN stimulation may reduce medication needs, which can lessen side effects such as dyskinesia and medication-related dystonia. |
| **Ventral intermediate nucleus (VIM)** | **Reduces:**<br>• Tremor |

> It's been six years since my DBS surgery. My medications have increased, but I'm still taking less than before DBS.
>
> ***— Vivek, person with PD***




# About the DBS System

## Neurostimulator (or implantable pulse generator)

The neurostimulator is the control center of the DBS system. It sends signals to the brain based on individualized programmed settings.

The neurostimulator:

- Is placed under the skin, usually near the collarbone
- Stores the settings that control how stimulation is delivered
- Sends signals through leads
- Comes with rechargeable or non-rechargeable batteries

### Battery Life and Replacement

Non-rechargeable batteries typically last 3 to 5 years. Rechargeable batteries can last up to 15 years but need regular charging. When the battery runs low, the neurostimulator is replaced during a short outpatient procedure. Your DBS team monitors battery levels at each visit, and you can also check with your controller.

## Leads

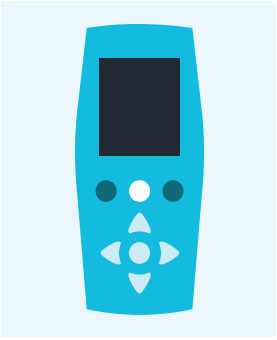
Leads are thin wires implanted in the brain that carry electrical signals from the neurostimulator to the brain. If symptoms affect both sides of the body, two leads — one on each side of the brain — are typically used. Each lead:

- Has **4 to 16 electrodes** (metal contacts) that direct electrical pulses to targeted areas.
- Uses **directional stimulation**, which means signals can be aimed at specific brain areas while avoiding others. This helps treat symptoms and avoid unwanted side effects.
- Connects to an **extension wire** that runs under the skin to the neurostimulator.

## Controller

The DBS team provides a handheld device or app for a phone or tablet that the person with DBS uses to control several key functions:

- Turn the neurostimulator on or off
- Check battery status
- Make limited adjustments as programmed by the DBS team
- Participate in remote programming sessions, depending on the system



## How the System Works

The system is turned on, typically a few weeks after surgery. It sends electrical signals to help regulate brain activity linked to symptoms.

- The programming team makes initial adjustments with a specialized device.
- They continue to adjust settings over time as symptoms change.

## Other Device Features

Some DBS systems offer extra features, each with possible benefits and considerations:

### Adaptive DBS

Unlike standard DBS systems with fixed settings, adaptive DBS (aDBS) also adjusts stimulation in response to brain activity.

- **Potential benefit:** It may help personalize stimulation based on real-time data.
- **Consideration:** Some people may not have brain signals that are strong or consistent enough to guide the adaptive feature.

### Image-Guided Programming

Standard DBS programming involves observing symptoms and side effects to guide adjustments. Image-guided programming uses brain scans to show lead placement and stimulation areas.

- **Potential benefit:** It may make stimulation adjustments faster and easier.
- **Consideration:** It requires specialized imaging tools and coordination between surgical and programming teams, which may limit availability.

### Remote Programming

Some DBS systems let the programming team adjust settings from a distance. The programmer connects using a controller or smartphone app during a scheduled session.

- **Potential benefit:** It may allow quicker access to stimulation adjustments and reduce the need to travel.
- **Consideration:** It relies on a strong internet connection and may not be offered at all clinics.

# Choosing a DBS System

There are a number of FDA-approved DBS systems for treating Parkinson's symptoms, made by Abbott, Boston Scientific and Medtronic. While each system has unique features, they all function in similar ways. One important factor in choosing a system is your DBS team's experience with the technology, as this can make programming and follow-up care easier.

Your DBS team will work with you to select a system that fits your needs and is available at your center. They will consider factors such as:

- Your symptoms and treatment goals
- Your daily routines and lifestyle
- The support you have at home

Be sure to bring up these topics with your care team if they haven't been discussed.

## Learn More

You can explore each system's features and support tools online:

- Abbott — AbbottDBS.com
- Boston Scientific — DBSandMe.com
- Medtronic — Medtronic.com/DBS

## Q How do I know if I am a good candidate for DBS?

A First, consult a movement disorders specialist — a neurologist with expertise in Parkinson's and related conditions — for a thorough evaluation of your symptoms, medications and health history. DBS is usually not recommended for atypical forms of Parkinson's.

**DBS may be an option if you:**

- ☐ Have a diagnosis of Parkinson's confirmed by a specialist
- ☐ Experience some improvement in symptom management with levodopa medication
- ☐ Have tried medication adjustments but still experience "off" time or side effects that affect daily life
- ☐ Have symptoms such as tremor or dystonia (painful cramping) that don't get better with medication
- ☐ Do not have serious mood, mental health or cognitive (thinking) problems
- ☐ Are healthy enough for surgery
- ☐ Have a support network who can help during surgery recovery and after
- ☐ Understand what DBS can and cannot do

## Q Can I wait before considering DBS?

A DBS is a personal decision, and not everyone with PD will need it or be a good candidate. If medications aren't managing your symptoms well, getting DBS earlier might make you feel better and improve your overall health. Waiting too long could increase surgery risks or, in some cases, make you ineligible later. Regular check-ins with your neurologist can help you understand your options.

**Q Who will be on my DBS care team?**

**A** Your team will include a neurologist who specializes in movement disorders and a neurosurgeon. Depending on your needs, other professionals such as nurses, physician assistants, neuropsychologists or rehab specialists may also be involved. A team approach helps support safe, effective care throughout the DBS process.

**Q How soon does DBS start working?**

**A** Some symptoms, such as tremor, may improve right away. Others can take days or weeks. The DBS team will adjust the stimulation settings over time to find what works best. Most people notice steadier results within about six months after surgery.

**Q Will I feel electricity from the DBS device?**

**A** People usually don't feel anything during daily use. You might notice tingling or dizziness when the device is turned on or adjusted, but it usually goes away quickly.

**Q Will I be able to stop taking Parkinson's medications after DBS?**

**A** Some people can reduce their PD medication doses after DBS, especially for tremor or dyskinesia (extra, involuntary movements). But most continue taking some medication, as symptoms will continue to change over time. Managing Parkinson's usually involves a combination of treatments — including medication, exercise and other therapies — tailored to each person's needs.

**Q Can I still try new treatments with DBS?**

**A** With DBS you can try any new FDA-approved treatment for Parkinson's symptoms. However, most clinical trials exclude people who have had DBS.

**Q Are there activities I should avoid after DBS?**

**A** Some movements may be restricted after surgery to protect the device. Your DBS team will tell you what to avoid and when it's safe to resume activities. You can usually start exercising 4–6 weeks after surgery unless advised otherwise. Always follow your team's guidance.

**Q Will my neurostimulator activate the metal detector at the airport?**

**A** It might, but it won't affect the device. Most airports let people with implants skip the detector. Bring your DBS card (which identifies your neurostimulator to security). If you've lost it, ask your DBS center for a replacement from the manufacturer.

**Q Can I have the DBS system removed?**

**A** Yes, the leads and neurostimulator can be removed. If the wires aren't causing issues, they may be left in to avoid surgery risks.

**Q Will my insurance cover DBS costs?**

**A** Since DBS is FDA-approved for Parkinson's, Medicare, Medicaid and most private insurance plans typically cover surgery and hardware costs. However, out-of-pocket expenses may apply depending on your plan. Most DBS centers can assist with insurance verification.

**Q What if DBS doesn't work?**

**A** DBS failure can happen, but most causes — such as issues with lead placement, hardware or programming — can often be fixed. DBS can also fail if someone isn't a good candidate for surgery.

FALL RISK

# 4

# Before, During & After DBS Surgery

If you decide to have deep brain stimulation (DBS), the next step is planning and preparing for your surgery and recovery.

Having a strong support network of family and friends is key to ensuring you have the care and resources you need throughout your DBS journey.

**In this chapter, you'll learn about:**

- What to expect during pre-surgery appointments and evaluations
- How to prepare for surgery and what to bring to the hospital
- Key aspects of post-surgery care and what to expect during recovery

## Once you have chosen a DBS center:

- **Prepare questions ahead of time.**
  Bring your concerns to appointments and make sure they're addressed. Feeling confident about the procedure is key. You might also find it helpful to speak with others who've had DBS.
- **Bring a family member or friend to appointments.**
  Someone you trust can help you absorb information, ask questions and offer support.
- **Meet all the members of your DBS team.**
  Understanding each person's role helps you know who to turn to at different stages of the process.

# Questions to Ask Your DBS Team

The following are suggested questions for your DBS team. Bring these with you to your pre-surgery appointments and write down answers to each question as you ask them.

**Who will be supporting me throughout the DBS process, and how do they communicate with each other?**

A coordinated team approach is essential for successful DBS surgery and follow-up.

**Do I have a choice between different devices?**

Some DBS centers work primarily with one device manufacturer, while others offer options. Your team will guide you based on your needs and preferences.

**Do I need DBS on both sides of my brain?**

Stimulation on both sides is common. Both leads are usually placed during the same surgery unless there's a medical reason to do them separately. In that case, the surgeries are spaced weeks or months apart.

**Will the neurostimulator be implanted at the same time as the brain lead, or will it be done later?**

Some centers do both at once, while others may place the neurostimulator in a separate procedure days or weeks later.

**After surgery, how long until I go home?**

Most people stay one night in the hospital. If confusion or other symptoms occur, your team may recommend a longer stay to ensure you're safe and ready.

**Will I be awake or asleep for the procedure?**

It varies by center. Ask your team about their approach — and why.

**Who will manage my DBS and medications after surgery?**

The DBS center usually handles programming at first. Later, your regular care team may take over. Ask how this will work in your case.

# Pre-Surgery Tests and Evaluations

Before surgery, you'll have several evaluations — some specific to DBS, others standard for any procedure.

These help your DBS team understand your symptoms, check for mental health concerns and ensure you will have support at home during recovery.

## Baseline Evaluation

- You'll visit the DBS center a few times. Your symptoms will be checked using a rating scale while you're on and off medication. This helps show which symptoms DBS is most likely to improve.

## Memory and Thinking Tests

- You'll take a set of tests, called a neuropsychological battery, to look for confusion, memory problems or mood issues. These are usually done by a neuropsychologist who works with people with PD. You may also see a psychiatrist if needed.

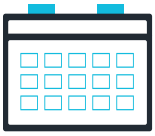
## Support Network Assessment

- A social worker will ask about your support system, including family and friends, to make sure you will have help before and after surgery. It is very important to have a care partner or someone with you full-time during recovery.

# Preparing for the Surgery

It is common to feel a mix of emotions as surgery approaches — excitement, worry or both. Talking with others who have had DBS can be reassuring, and having a clear plan can make the process feel more manageable.

Use this checklist to help you prepare:



## A Few Weeks Before Surgery

- ☐ **Know where to go on surgery day.** Write down the address, phone numbers and parking instructions for the center. Consider doing a practice trip.
- ☐ **If traveling, arrive a day early and stay nearby.** Ask the DBS center about lodging options.
- ☐ **Confirm medication timing with your DBS team.** Most advise taking the last dose the night before and skipping medication on surgery day.

## Packing List for the Hospital Stay

Bring the following items:

- ☐ **Medications**, with a list of names, dosages and times. Pack at least a week's worth in the original bottles.
- ☐ **Emergency contact information.** Bring phone numbers for your care partner and at least one other person the hospital can reach if needed.
- ☐ **Parkinson's Foundation Hospital Safety Guide**, which includes practical tools such as medication forms to help you communicate your needs.

  ▶ **Download or order your free copy** at Parkinson.org/HospitalSafety or call the Helpline at 1-800-4PD-INFO (1-800-473-4636).

### The Night Before Surgery

- ☐ Review the instructions from your DBS team, including when to stop eating and drinking before surgery.
- ☐ Try to get a good night's sleep.

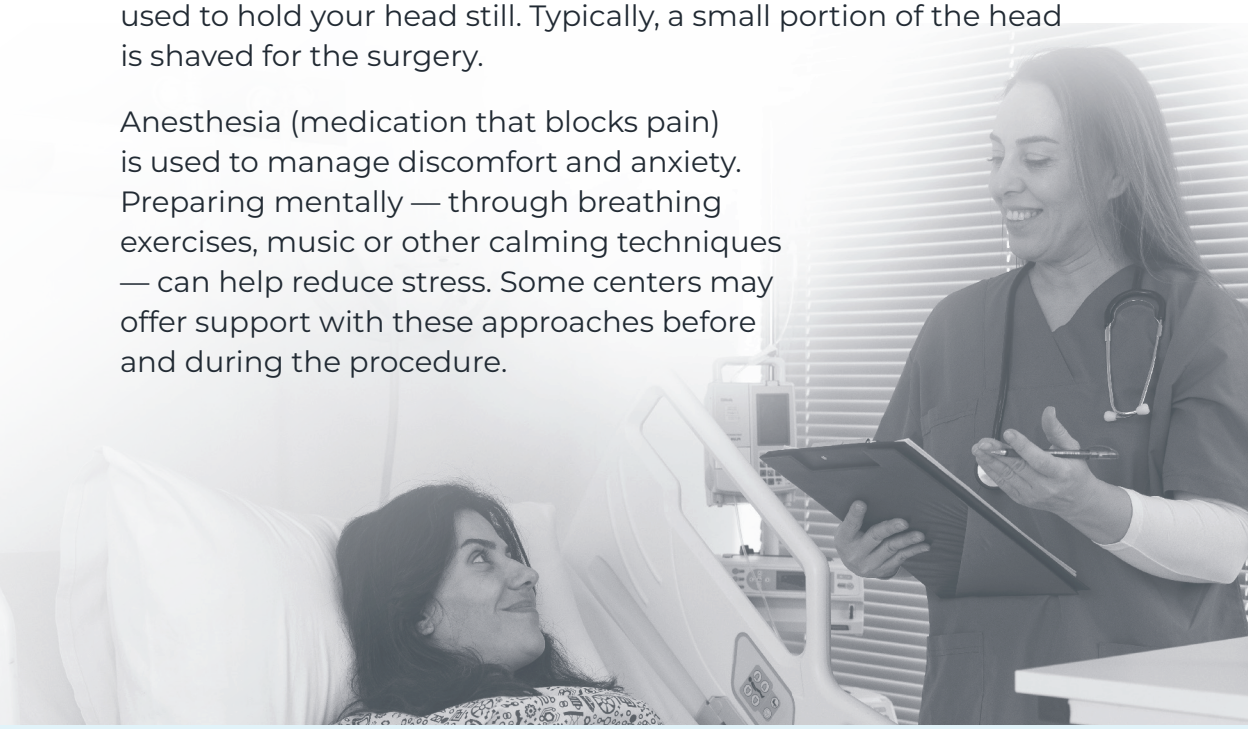
## The Surgery

Surgery can vary from one DBS center to another. It may be performed while you're awake or asleep, depending on the center's approach. Some neurosurgeons use robotic tools, some use a headframe to stabilize the head, and others use a frameless method. The best techniques are the ones your team knows well and uses most often.

### Preparing for Surgery

Before surgery, high-resolution imaging (MRI or CT scans) helps locate the target area in the brain. A frame or screws may be used to hold your head still. Typically, a small portion of the head is shaved for the surgery.

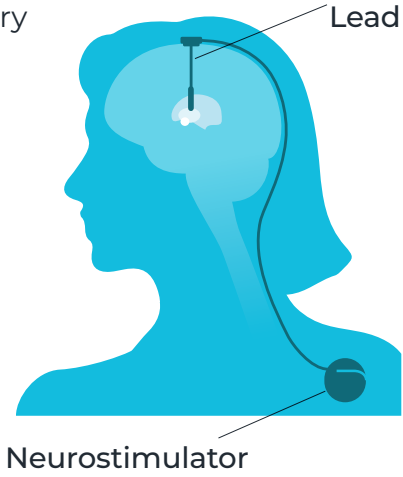
Anesthesia (medication that blocks pain) is used to manage discomfort and anxiety. Preparing mentally — through breathing exercises, music or other calming techniques — can help reduce stress. Some centers may offer support with these approaches before and during the procedure.



## Lead Placement

Leads are placed in the targeted brain areas through small openings in the skull. The lead placement surgery involves an overnight hospital stay for monitoring. There are two approaches your neurosurgeon can use to place your DBS lead(s):

- **Microelectrode recording-guided placement:** MRI or CT scans are used to plan surgery and check placement of the lead by listening to signals from your brain cells using microelectrode recording.
- **MRI or CT image-guided placement:** MRI or CT scans are used to plan the surgery and guide the lead's placement in real time. This method can be done while you're asleep under anesthesia.




## Neurostimulator Placement

The neurostimulator is usually placed under the collarbone. In some cases, it may be placed in the abdomen or another location. The device may be placed in one of two ways:

- **At the time of the lead placement surgery:** This typically doesn't extend the hospital stay.
- **In a separate procedure a few weeks later:** This is usually a same-day procedure, and you'll go home the same day.

Since the stimulator is not turned on immediately, you will continue taking your Parkinson's medication as prescribed. Some people notice an improvement in symptoms before the stimulator is activated, but this is temporary.

The stimulator will be activated and programmed after you have had time to recover from surgery.

> I didn't feel like I had brain surgery. I was completely comfortable. There was no pain until I got home — and even then, it was just when the staples rubbed against something.
>
> ***— Lisa, person with PD***

## Care After Surgery

When you leave the hospital, the DBS team will give you instructions on:

- **Incision (surgical cuts) care:** This includes how to clean the area, monitor for infection and what to expect as the wounds heal.
- **Follow-up appointments:** You'll be given dates for follow-up visits to remove stitches or staples, place the neurostimulator (if not done during the brain surgery) and program the neurostimulator.
- **Parkinson's medication:** You will receive guidance on when and how to take your PD medication after surgery, including any changes or adjustments.

These instructions will also include emergency phone numbers in case you have any concerns or need assistance.

Side effects or complications might develop in the first few weeks or even months after surgery. Read the next chapter, **DBS Risks & Side Effects,** to learn what signs to watch for.

# 5

# DBS Risks & Side Effects

Deep brain stimulation (DBS) involves at least two surgeries — one to place the lead or leads in the brain, and another to implant the neurostimulator under the skin in the chest.

As with any surgery, DBS carries risks. While most people recover well and benefit from DBS, some may experience temporary side effects, device issues or — rarely — serious complications.

**In this chapter, you'll learn about:**

- Surgical and recovery-related risks
- Stimulation side effects
- Device complications
- Serious or lasting complications

**These sections outline common DBS risks, complications and side effects. For a full list, consult your DBS team.**

# Risks During Surgery and Recovery

Complications can occur during surgery or in the days afterward, such as:

- **Pain, numbness, swelling or inflammation at the surgery sites:** These are common but usually temporary. Worsening symptoms may signal infection.
- **Infections in the brain, scalp or chest area:** These mostly affect the tissue around the implanted device in the scalp or chest; brain infections are very rare.
- **Allergic reactions to implanted materials:** Share any known allergies to metals or implants with your DBS team early on.
- **Confusion:** Temporary disorientation may occur after surgery, especially in older adults. This usually improves within a few days.
- **Seizures:** While rare, seizures are possible during or shortly after surgery — usually within 24 hours, sometimes within a week.

# Stimulation Side Effects

DBS can cause side effects. Most are short term and often improve when medication or stimulation settings are adjusted.

## Short-Term Stimulation Effects

These may occur early on but usually go away within days or weeks:

- **Sensory effects** may include headache, dizziness, tingling or electric shock-like sensations.
- **Mood, memory and thinking changes** can be caused by changes in stimulation or when medication is reduced too much after surgery.

### Possible Long-Term Stimulation Effects

Some side effects may last longer or come and go. These often affect movement or speech and include:

- Slurred speech
- Trouble with balance
- Dyskinesia (extra, involuntary movements)
- Muscle pulling or tightness
- Trouble walking, including freezing (feeling stuck in place)
- Problems with coordination

Many of these improve with stimulation or medication adjustments. Physical, occupational or speech therapy may also help with walking, balance, coordination or communication.

## Device-Related Complications

Problems with the implanted DBS system are uncommon but can interfere with stimulation or cause discomfort. These may include:

- Wires moving, breaking or placed in the wrong spot
- The neurostimulator not working or getting damaged
- Skin irritation near the device

Some cases may need further testing or surgery.

## Serious or Permanent Complications

These events are rare but possible, including:

- **Stroke** occurs in 1 to 3% of DBS surgeries (usually bleeding-related) and may cause weakness, numbness or speech and vision issues.
- **Death** is very rare — the risk is between 0.2 to 0.6%, depending on factors such as age, health and experience of the surgeon.

# 6

# DBS Programming

After deep brain stimulation (DBS) surgery, the device must be adjusted based on each person's symptoms. This process — called programming — involves turning on the device and identifying the stimulation settings that reduce symptoms while avoiding unwanted side effects.

Programming typically begins soon after surgery and continues through a series of follow-up visits. These sessions help fine-tune the settings as symptoms and treatment goals evolve.

**In this chapter, you'll learn about:**

- How DBS programming works
- What to expect at the initial visit
- How settings are adjusted over time

# Programming Overview

Programming is an ongoing process that adjusts the DBS system to meet your specific needs. In each session, a trained member of your DBS team — often a neurologist, a registered nurse or a physician assistant — uses a tablet-based system (sometimes called a clinician programmer) to change the stimulation settings and observe how your symptoms respond. They may direct stimulation to different areas of the brain or adjust its strength and speed to achieve the best results with the fewest side effects.

Some devices streamline programming through image-guided approaches based on MRI or CT scans or adaptive systems that respond to real-time brain signals. Programming is often in person, though some devices support remote sessions depending on your DBS team.

Programming is not a one-time visit — it usually takes multiple sessions to find the most effective settings. Adjustments continue over time as your symptoms change.

## Medications:

- Most people continue taking their Parkinson's medications after DBS, but at lower doses.
- Your team may ask you to hold medication before a visit to make symptoms easier to evaluate.
- Bring your medications to each session in case you need to take them during or after programming.

# Initial DBS Programming

The first programming visit usually happens a few weeks after surgery, once brain swelling has gone down. In some cases, the DBS device may be turned on before you leave the hospital.

## Preparing for the Visit

- Your team will guide you through what to expect. You may receive instructions before your visit — how to prepare, what to bring or whether to pause medications. If anything is unclear, ask your DBS team. Some people bring a care partner to help describe symptoms or take notes. It can also help to jot down any changes you've noticed since surgery to share during the visit.

## Finding the Right Settings

- At this visit, your DBS team will turn on the device and begin adjusting settings to find a range — called the therapeutic window — where stimulation improves symptoms with minimal side effects. They'll test different configurations to find what works best for you.

  You may be asked to move, speak or perform simple tasks so your team can observe how stimulation affects your symptoms. Some symptoms, such as tremor, may improve quickly. Others may take hours or days to respond.

## Possible Side Effects

- Side effects — such as tingling or speech changes — may appear during the visit or later, sometimes after a dose of medication. Other temporary side effects may include balance issues, confusion, increased dyskinesia or changes in mood or personality.

  Report any stimulation side effects to your team — adjustments can help, and many fade over time.

# Ongoing DBS Programming

After the initial programming session, you will have additional appointments. The number and timing of these can vary depending on the center. Some schedule check-ins every one to two months during the first six months, while others may space them out more.

After therapy stabilizes, many people only need one or two follow-ups per year to adjust settings or check the device.

Key aspects of ongoing programming include:

- **Adjusting settings:** Your symptoms may change over time, so the DBS team will adjust the settings as needed to improve symptom relief and minimize side effects.
- **Managing medications:** You will likely continue taking Parkinson's medications, though your doses might be adjusted as the DBS system is fine-tuned.
- **Battery checks:** The team will check the battery status at each visit. As the battery gets low, your symptoms may worsen. The team will plan for replacement before this happens.

> I can move my right side much better than before having DBS and I am taking less medicine now.
>
> I did have some trouble with tingling on the right side of my face, and with slurred speech, so they had to change the settings a couple of times.
>
> After a few programming sessions, the numbness and speech were better.
>
> ***— Mike, person with PD***

# More Programming Information

After surgery, your DBS team will give you a way to interact with your system — either a handheld controller or a downloadable app on your smartphone, depending on the manufacturer.

These tools let you:

- Turn the DBS device on or off
- Toggle between preprogrammed settings
- Increase or decrease settings within a programmed range
- Check the battery level
- Join remote programming sessions, if supported by your device and DBS team

# 7

# Focused Ultrasound

Focused ultrasound is a procedure used to treat movement symptoms in Parkinson's disease (PD). Like deep brain stimulation (DBS), it targets specific brain areas involved in movement. Rather than using electrical signals to regulate brain activity, focused ultrasound uses sound waves to heat and destroy a small area of brain tissue. This creates a lesion — a small spot of changed tissue — that helps interrupt abnormal signals.

Focused ultrasound is an advanced treatment option for people with PD when medications no longer control symptoms consistently or cause significant side effects.

**In this chapter, you'll learn about:**

- How focused ultrasound works
- Other types of lesion therapy
- Risks and considerations of lesion therapy
- Focused ultrasound vs. DBS

# Overview of Focused Ultrasound

Focused ultrasound (FUS) is a type of lesion therapy that can reduce movement symptoms such as tremor, stiffness and slowness. The U.S. Food and Drug Administration (FDA) approved treatment on one side of the brain for Parkinson's tremor in 2018. Over time, approval expanded to other movement symptoms. In 2025, the FDA approved treatment on both sides of the brain, with procedures spaced at least six months apart to lower risk of side effects.

## How It Works

FUS uses sound waves to create small burns (lesions) in areas of the brain with abnormal activity. These burns help reduce movement symptoms by changing how signals move through those areas.

### During the procedure:

- The head is shaved, and a helmet is positioned to direct the ultrasound waves to the target area.
- Chilled water flows through the helmet to keep the head cool.
- MRI scans guide the treatment in real time.
- No incisions or general anesthesia are needed, which makes it less invasive than traditional surgery.

# Other Types of Lesion Therapy

Other lesion therapies also target parts of the brain to help with movement symptoms in Parkinson's. The main types include:

- **Radiofrequency surgery:** A probe heats and destroys brain tissue through a small opening in the skull. One side is left untreated to lower side-effect risk, so symptoms on that side may continue.
- **Gamma knife radiosurgery:** Uses focused radiation to create lesions without cuts. It is rarely used because there are limits to how much radiation can be safely given.

# Areas of the Brain Treated with Lesion Therapy

Lesion therapies target different brain areas based on a person's symptoms and treatment goals.

- **Pallidotomy** targets the globus pallidus internus. It can help with slow movement, stiffness and dystonia. It may also reduce tremor, "off" time and dyskinesia (extra, involuntary movements caused by levodopa).
- **Thalamotomy** targets the thalamus to reduce tremor.
- **Subthalamotomy** targets the subthalamic nucleus. It may help with slowness and stiffness, similar to pallidotomy, but has a higher risk of causing dyskinesia. Subthalamotomy remains experimental for both focused ultrasound and radiofrequency surgery and is not FDA approved.

# Lesion Therapy Risks and Warnings

Lesion procedures involve several risks. These include:

- Stroke, bleeding, seizures, weakness, numbness, vision or speech changes, balance issues and stroke-like symptoms
- Higher risk of side effects when both sides of the brain are treated
- Rare but possible long-term effects, even with treatment on one side

**Important to know:** Lesions are permanent and cannot be adjusted if side effects occur.

# Focused Ultrasound (FUS) vs. Deep Brain Stimulation (DBS)

| Feature | FUS | DBS |
|---|---|---|
| **Ability to adjust** | Permanent brain changes; no way to modify if side effects occur or results are not ideal | Reversible; allows ongoing adjustments to improve symptom management over time |
| **Ongoing maintenance** | No programming sessions or battery replacements | Regular programming adjustments and occasional battery replacements |
| **Candidate restrictions** | No surgery or general anesthesia, sometimes safer for people with health concerns | Involves surgery and general anesthesia; candidates must be in good health |

# Appendix

## Key Safety Considerations for DBS & Pump Medications

### Hospital Safety

People with Parkinson's disease (PD) often face extra challenges during hospital stays. Delayed, missed or changed medications — along with limited movement — can lead to complications, worsening symptoms and slower recovery.

The **Hospital Safety Guide** helps you prepare for a hospital stay and outlines steps to follow while you're there. It also includes forms and guidance for your care team. Download or order your copy at Parkinson.org/HospitalSafety or contact our Helpline at 1-800-4PD-INFO (1-800-473-4636) for assistance.

If you use a DBS device or pump medication, such as Duopa, Vyalev or Onapgo, follow these extra steps before a planned or emergency hospital visit.

- Fill out the **Device-Specific Needs** form in the Hospital Safety Guide to share important information about your device or medication with the hospital care team.
- Be ready to bring your own medication supplies in case the hospital pharmacy doesn't stock them.

## Pump Medication Safety

This section outlines general safety guidelines for therapies that use a pump to deliver medication. Always confirm details with your care team or device manufacturer before proceeding. For manufacturer support, see page 19.

### MRI Guidelines

If you use a pump to deliver medication, special precautions are needed for MRI:

- Stop and disconnect the pump before the scan. Store it outside the MRI room.
- Start oral medications if the Vyalev or Onapgo are disconnected for more than one hour, or Duopa for more than two hours.
- For Vyalev and Onapgo, replace the cannula and pump tubing if the interruption lasts more than one hour.

Note: Duopa pumps may interfere with ECG readings.

### Infection Prevention

To reduce the risk of infection:

- Use sterile techniques when handling the pump, tubing or infusion site. Wash hands and avoid unnecessary contact with the insertion site or equipment.
- Clean the infusion or stoma site daily with antiseptic wipes or as directed by your care team.
- Check the site for redness, swelling, warmth, pain or drainage.

If you notice any of these signs — or develop fever, chills or feel unwell — contact your care team promptly.

## DBS Safety

The following guidance highlights common procedures and general DBS safety considerations. Always confirm with your DBS care team and device manufacturer before proceeding.

### Safe Imaging and Procedures

- **CT scans, DAT/SPECT scans, X-rays, fluoroscopy and PET scans** are generally safe.
- **Ultrasound** (carotid or abdominal) is safe.
- **Radiation therapy** is safe with shielding over the neurostimulator.
- **Mammograms** are safe if pressure doesn't compress the device.
- **Pacemakers** may be implanted if placed at least 10 inches from the neurostimulator.

### MRI Guidelines

- MRI should only be performed by staff trained in DBS imaging.
- Avoid MRI if the neurostimulator is not in the chest or abdomen, or if there is hardware damage.
- Some devices must be set to 0.0 volts before an MRI.
- Turning off the DBS device may improve EKG or EEG readings.

### Precautions and Restrictions

- Avoid procedures directly over the neurostimulator or wires.
- **Electrocautery** (heat tool) may be used during surgery but only with special precautions. Consult your DBS team for guidance.
- Use **lithotripsy** (kidney stone treatment) only if essential, with the neurostimulator turned off and protected.
- Avoid **diathermy** (electric heat therapy), as it may interfere with the device.

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## About the Parkinson's Foundation

The Parkinson's Foundation makes life better for people with Parkinson's disease by improving care and advancing research toward a cure.

In everything we do, we build on the energy, experience and passion of our global Parkinson's community. Information about Parkinson's and resources are available at Parkinson.org.

**Your generosity makes this publication possible.**
The Parkinson's Foundation is proud to provide this guide and other educational materials at no cost to people around the world. If you found this resource helpful, please consider a donation to help us continue to offer up-to-date, high-quality educational materials about Parkinson's, as well as fund innovative research and support services.

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